

Advanced Public Economics — Applied Tutorial

Session VII: Tax Reforms

Daniel Weishaar

Roadmap for Session VII

- ① Recap lecture formulas on the **revenue and welfare implications of tax reforms**.
- ② Analyze (small) reforms of the status quo tax and transfer system in Germany.
 - 2.1 Pareto-improving reforms.
 - 2.2 Welfare-improving reforms.

Recap from Lecture — Revenue Functions

Under quasi-linear preferences and iso-elastic effort costs, revenue function

$$\mathcal{R}(y) = \underbrace{1 - F_y(y)}_{?} - \underbrace{\varepsilon y f_y(y) \frac{T'(y)}{1 - T'(y)}}_{?}$$

gives the **revenue impact** of a small **increase of marginal tax rates** by τ in a small income bracket starting at income level y .

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⇒ What about extensive margin responses?

Recap from Lecture — Revenue Functions — Extensive Margin (I)

The formula under intensive margin responses can be extended — requires some **fixed cost of working** in model in addition to variable effort costs.

$$\mathcal{R}_{int}(y) = \underbrace{1 - F_y(y)}_{\text{Mechanical Effect}} - \underbrace{\varepsilon_y f_y(y) \frac{T'(y)}{1 - T'(y)}}_{\text{Behavioral Response, Int. Margin}}$$

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Interpretation:

- ▶ An increase in marginal tax rates at income level y leads to a higher tax liability for everyone with income above.
- ▶ For some of these individuals, the increase in tax liability means that they are no longer willing to pay the fixed cost of working, they stop working completely.

Recap from Lecture — Revenue Functions — Extensive Margin (II)

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- $\pi_0(y')$: Participation elasticity, i.e., what share of individuals leave the labor market when net income is decreased by one percent. $\Rightarrow$ As for ε , we make assumptions regarding π_0 .

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- ▶ $\pi_0(y')$: Participation elasticity, i.e., what share of individuals leave the labor market when net income is decreased by one percent. $\Rightarrow$ As for ε , we make assumptions regarding π_0 .
- ▶ $t_p(y')$: Participation tax rate, i.e., change in net tax liability when individual starts to work. $\Rightarrow$ Can be estimated from effective tax schedule.

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We have already estimated most of the **ingredients** for the estimation of **revenue functions for Germany**:

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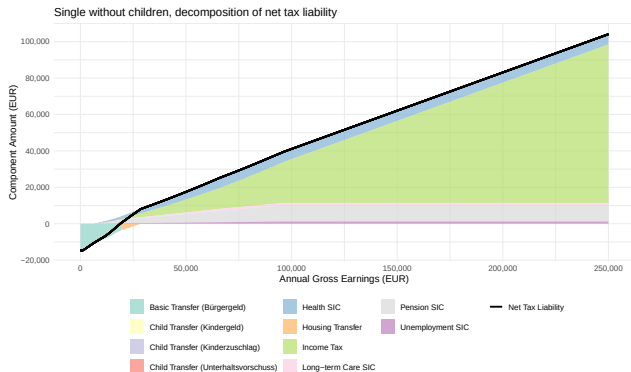
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⇒ *Note*: For single household with children, assume children of age 9 and age 5, rent of 650 EUR and 130 EUR heating costs. Other ingredients can be taken over from the single household without children.

Revenue Function Ingredients — Tax Liability and EMTR

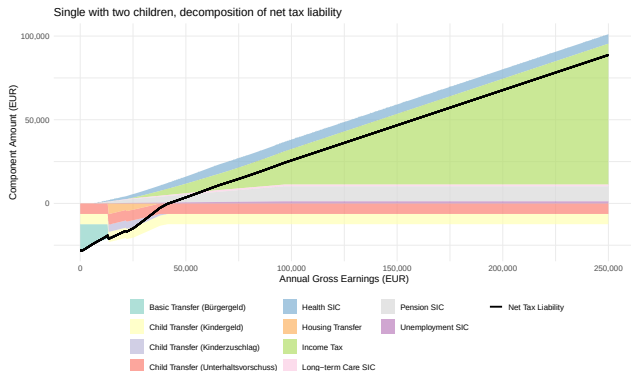
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Singles without Children:



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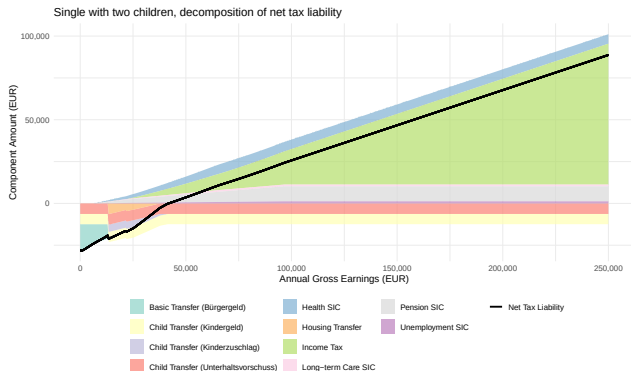
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- For singles with two children, similar picture for the net tax liability decomposition as for singles without children.

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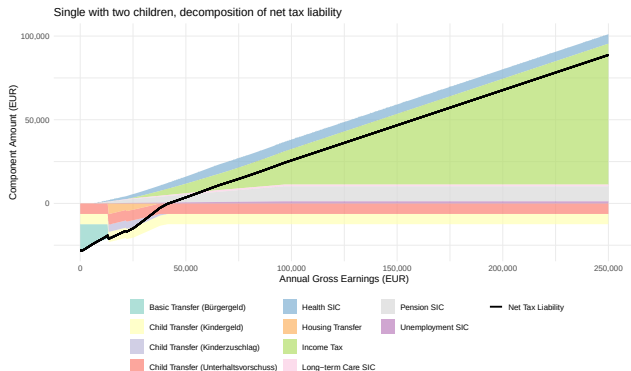
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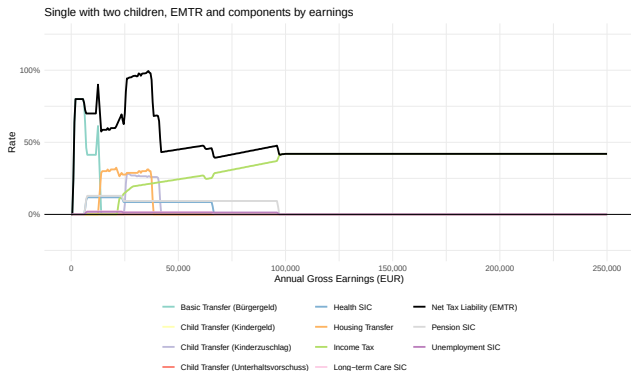
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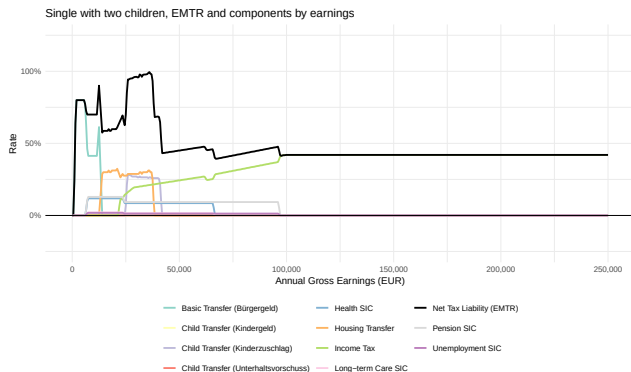
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- ▶ EMTR schedule features *kinks* — discrete changes in the marginal tax rate.

Revenue Function Ingredients — Tax Liability and EMTR

Singles with two Children:



Question from last time: who administers the different transfers? [Details](#)

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Revenue Function Ingredients — Participation Tax Rate

The extensive margin requires an estimate of the **participation tax rate** t_p . It measures, how much the net tax liability changes when a person starts working.

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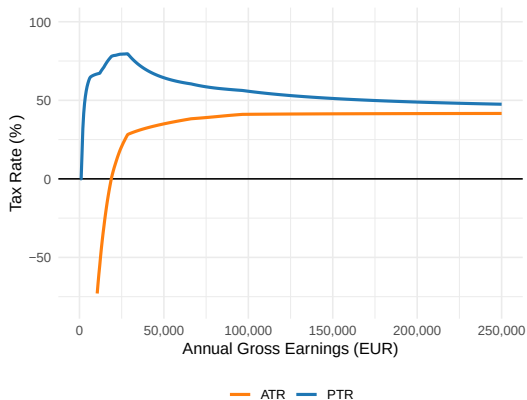
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⇒ **Hands-on exercise:** Calculate and plot the participation tax rate and average tax rate for singles with and without children in Germany.

Revenue Function Ingredients — Participation Tax Rate — Results

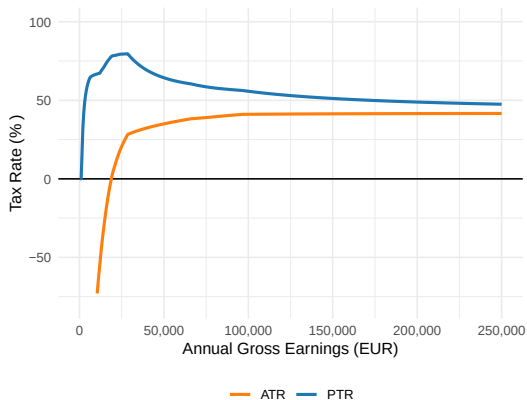
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- Average tax rate (ATR) can be negative due to transfer, participation tax rate (PTR) non-negative.

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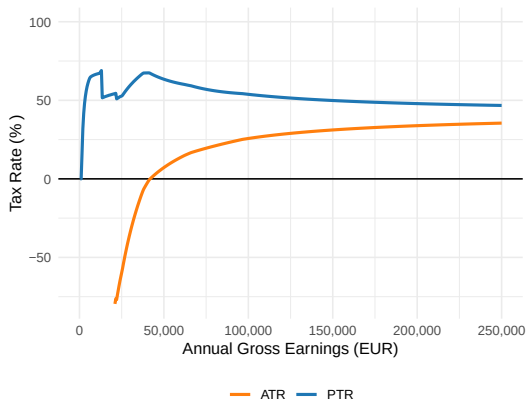
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Singles with two Children:



- ▶ Average tax rate (ATR) can be negative due to transfer, participation tax rate (PTR) non-negative.
- ▶ ATR monotonically increasing. This does not hold for PTR.
- ▶ For singles with children, notches imply discrete jumps in ATR and PTR.

Revenue Function — Estimation

We now have all ingredients to compute revenue functions (incl. extensive margin).

$$\mathcal{R}_{int+ext}(y) = \underbrace{1 - F_y(y)}_{\text{Mechanical Effect}} - \underbrace{\varepsilon y f_y(y) \frac{T'(y)}{1 - T'(y)}}_{\text{Behavioral Effect, Int. Margin}} - \underbrace{\int_y^\infty f_y(y') \pi_0(y') \frac{t_p(y')}{1 - t_p(y')} dy'}_{\text{Behavioral Effect, Ext. Margin}}$$

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► Effective marginal tax rates ✓

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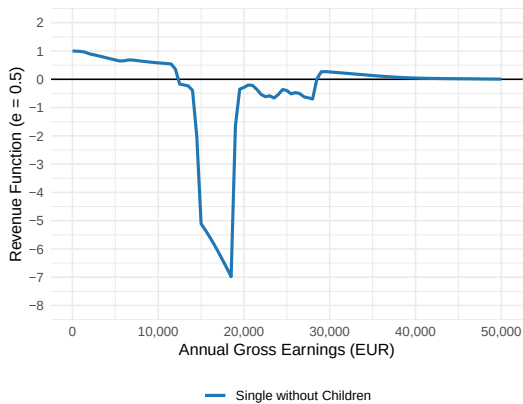
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⇒ **Hands-on exercise:** Estimate revenue functions, start with intensive margin only.

Revenue Function — Results — Intensive Margin Only (I)

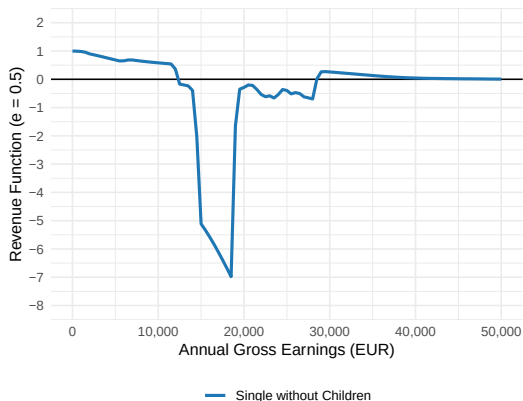
Singles without Children ($\varepsilon = 0.5$):



- Revenue function shows the revenue effect of a small MTR increase at a specific income level.

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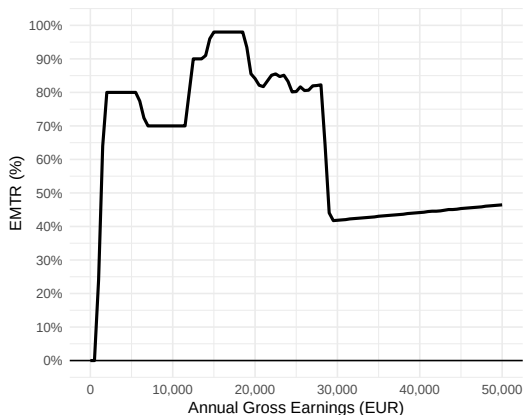
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 - ▶ Bierbrauer et al. (2023) provide necessary and sufficient conditions for Pareto efficiency related to revenue functions. [Details](#)
- ① **Negative values** indicate that raising tax rates comes with a loss in tax revenue. For $\varepsilon = 0.5$ the case for incomes between 12,000 EUR and 28,000 EUR $\Rightarrow$ **Inefficient!**

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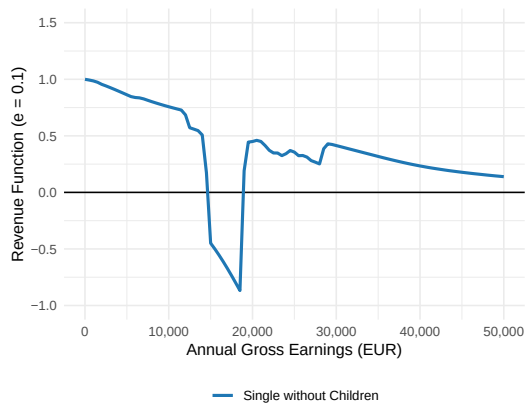
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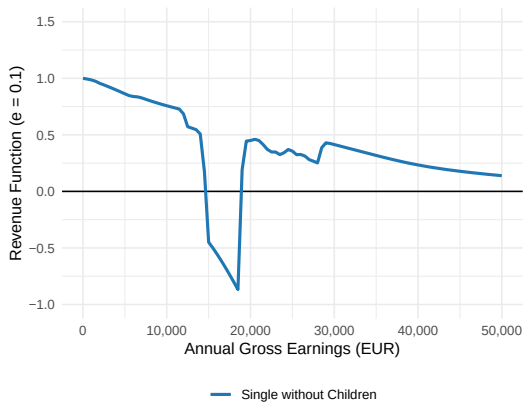
Revenue Function — Results — Intensive Margin Only (II)

Singles without Children ($\varepsilon = 0.1$):



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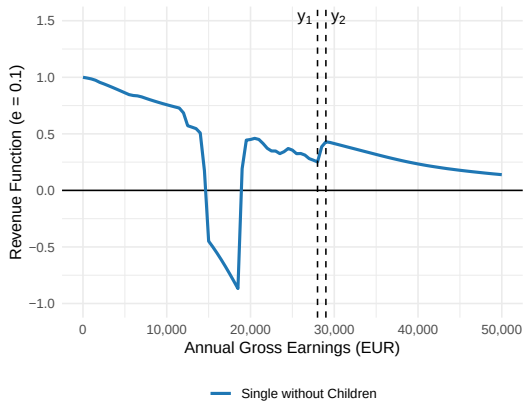
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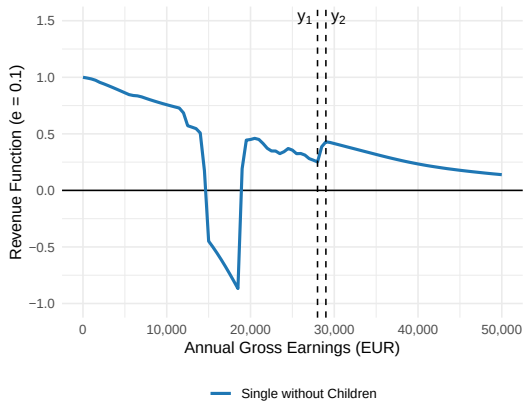


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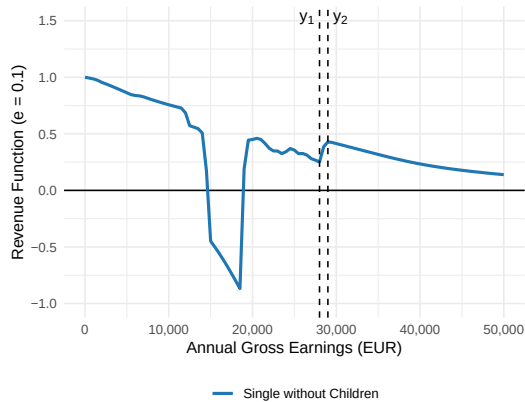


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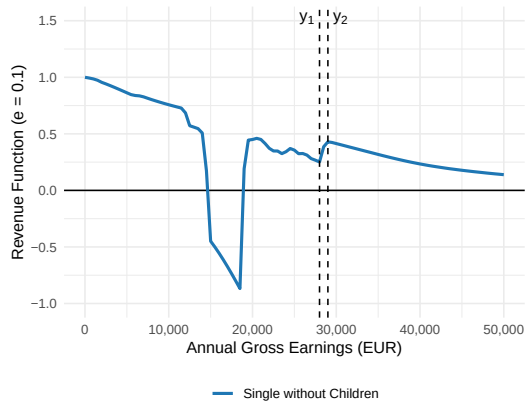


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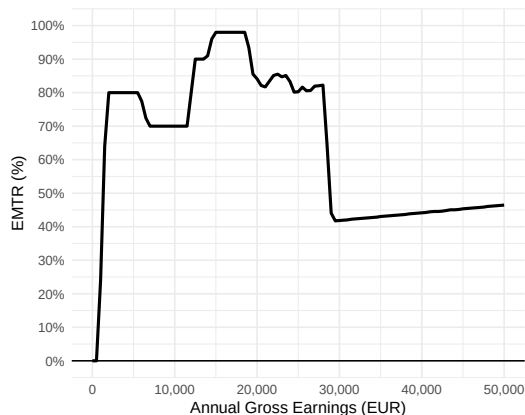


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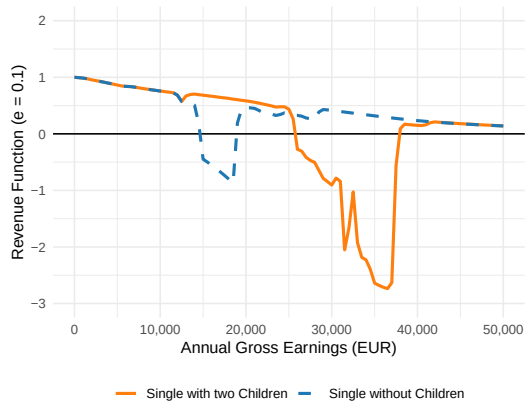


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 - At y_1 , the mechanical effect is larger than for y_2 , because more are affected by the tax hike. Still, revenue effect is smaller at y_1 .
 - Cure inefficiency by small decrease of MTR at y_1 and increase at y_2 .
- Revenue function jumps observed where EMTR drops. Also relevant outside of Germany (Hansen and Weishaar, 2025). [Details](#)

Revenue Function — Results — Intensive Margin Only (III)

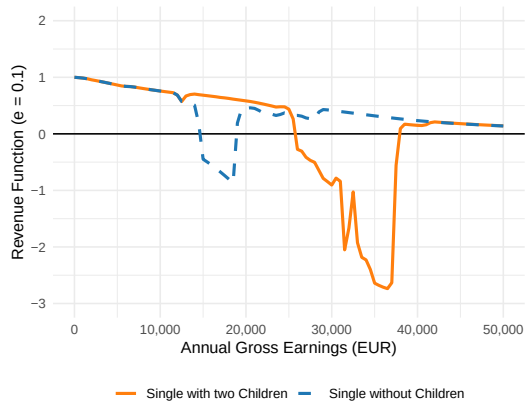
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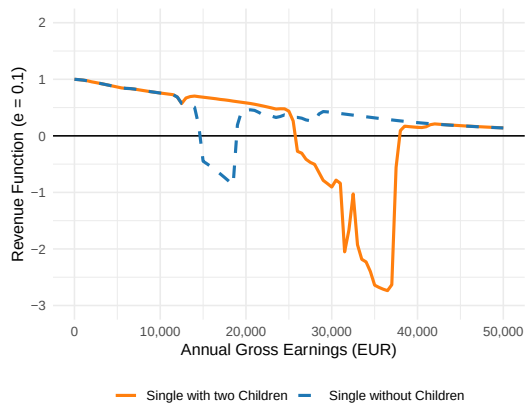
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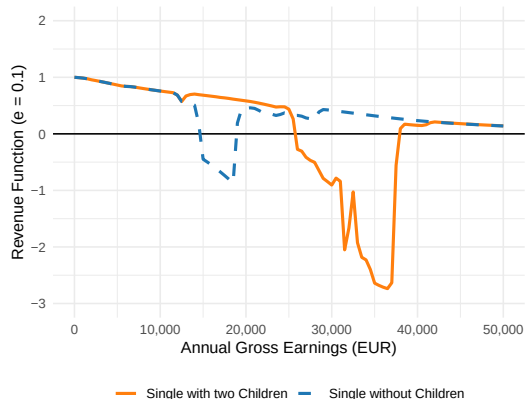


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Note of Caution: We have used here the general income distribution of singles. Ideally we would need to differentiate between singles with and without children and use the year corresponding to the tax system. We did not do so, since we only have access to SOEP training data.

Revenue Function — Results — Intensive Margin Only (III)

No Children vs. Children ($\varepsilon = 0.1$):

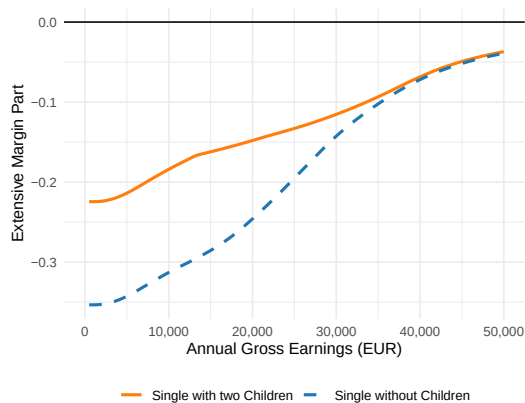


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Let's now include extensive margin responses. $\Rightarrow$ [Hands-on exercise](#)

Revenue Function — Results — Extensive Margin



$$\underbrace{- \int_y^\infty f_y(y') \pi_0(y') \frac{t_p(y')}{1 - t_p(y')} dy'}_{\text{Behavioral Effect, Ext. Margin}}$$

- Use numerical integration to obtain the extensive margin part.

Revenue Function — Results — Extensive Margin



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- Use numerical integration to obtain the extensive margin part.
- Extensive margin part shifts revenue function downward. Under extensive margin responses, a tax hike has lower revenue effects, because it triggers some people to stop working.

From Revenue Functions to Welfare Functions

Revenue functions describe the pure revenue effect of a local tax increase. A social planner might not always like to increase revenue through higher taxes, because she cares about those people from which these taxes are levied.

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Welfare Effect of Reforms (here under intensive margin responses):

$$\begin{aligned}\mathcal{W}(y) &= \mathcal{R}(y) - (1 - F_y(y)) \mathcal{G}(y) \\ &= (1 - F_y(y)) - \varepsilon_y y f_y(y) \frac{T'(y)}{1 - T'(y)} - (1 - F_y(y)) \mathcal{G}(y)\end{aligned}$$

where $\mathcal{G}(y)$ is the average welfare weight of individuals with incomes larger than y .

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Special Cases:

- ▶ Under equal weights, i.e., $g(y) = 1$, and for $T' > 0$: $\mathcal{W}(y) < 0$.
- ▶ Under Rawlsian weights, i.e., $g(y) = 0$ for $y > 0$: $\mathcal{W}(y) = \mathcal{R}(y)$.

Welfare Functions — Estimation

Some **Hands-on exercises**: Estimate the welfare effect of local tax increases for singles without children under the following hypothetical preferences of a social planner:

- ① **Working Poor**: concentrates all weight on singles with incomes of less than 30,000 EUR.
- ② **Inequality Averse**: assigns declining weights in a way that a doubling of incomes halves the welfare weight.
- ③ **Loves the Rich**: assigns linearly increasing weights.

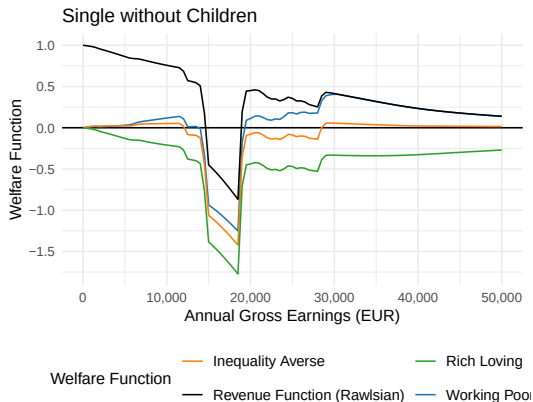
Note: Normalize all welfare weights to have a mean of 1.

Welfare Functions — Results



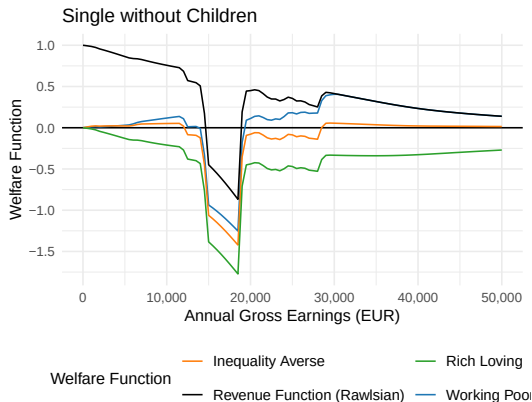
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Welfare Functions — Results



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- Welfare functions can agree or disagree to a tax hike at a specific income level y .

Welfare Functions — Results



- ▶ $\mathcal{G}(y)$ can take different forms, all $\mathcal{G}(0) = 1$.
- ▶ Welfare functions can agree or disagree to a tax hike at a specific income level y .
- ▶ Welfare functions always agree when tax system is inefficient (for positive welfare weights).

That's it! See you next time!

References

References I

-  BIERBRAUER, F. J., P. C. BOYER, and E. HANSEN (2023). “Pareto-Improving Tax Reforms and the Earned Income Tax Credit.” *Econometrica* 91 (4), pp. 1077–1103.
-  HANSEN, E. and D. WEISHAAR (2025). “Self-Financing Tax Cuts Around the World: New Theoretical Results and Applications to 33 Countries.” Mimeo.

Bierbrauer et al. (2023) — Sufficient Conditions

Bierbrauer et al. (2023) characterize Pareto efficiency in relation to revenue functions.

Theorem (Bierbrauer et al. (2023))

There is a Pareto-improving tax reform unless

- ▶ $\mathcal{R}_0(y) \in [0, 1]$ for all $y \geq 0$ and
 - ▶ $\mathcal{R}_0$ is monotonically decreasing for all $y \geq 0$.
- ▶ $\mathcal{R}_0(y) < 0 \rightarrow$ self-financing one bracket tax cut at y .

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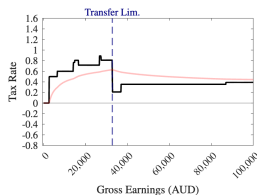
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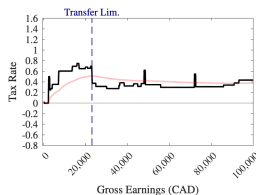
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 - ▶ $\mathcal{R}_0(y)$ **increasing** $\rightarrow$ self-financing **two-bracket tax cut** around y .
 - ▶ $(\mathcal{R}_0(y) > 1 \rightarrow$ self-financing **one bracket tax increase** at y .)

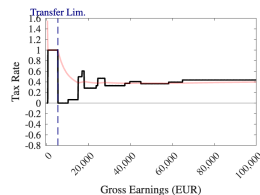
Hansen and Weishaar (2025) — Cross-Country Analysis — EMTR



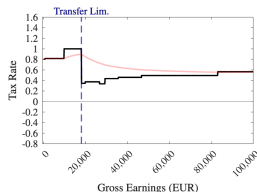
(a) Australia



(b) Canada



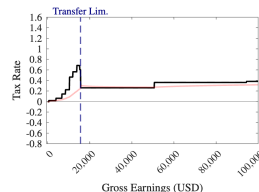
(c) Spain



(d) Finland



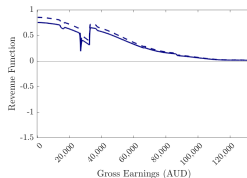
(e) UK



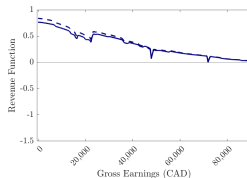
(f) USA

A lot of countries feature discrete MTR drops where transfers are phased out.

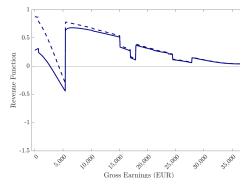
Hansen and Weishaar (2025) — Cross-Country Analysis — $\mathcal{R}$ -Function



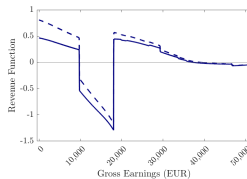
(a) Australia



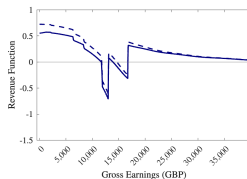
(b) Canada



(c) Spain



(d) Finland



(e) UK



(f) USA

Non-monotonicities indicate inefficiencies in all of the selected countries.

Who administers the different transfers?

Transfer	Executing Body	Executing Level	Main Ministry
Arbeitslosengeld I	Bundesagentur für Arbeit	Bund	Bundesministerium für Arbeit und Soziales
Bürgergeld (SGB II)	Jobcenter (Bundesagentur für Arbeit + Kommune)	Bund / Kommune	Bundesministerium für Arbeit und Soziales
Sozialhilfe (SGB XII)	Kommunale Sozialämter	Kommune	Bundesministerium für Arbeit und Soziales
Grundsicherung im Alter / bei Erwerbsminderung	Kommunale Sozialämter	Kommune	Bundesministerium für Arbeit und Soziales
Wohngeld	Kommunale Wohngeldstellen	Kommune	Bundesministerium für Wohnen, Stadtentwicklung und Bauwesen
Kindergeld	Familienkasse (bei der Bundesagentur für Arbeit)	Bund	Bundesministerium der Finanzen
Kinderzuschlag	Familienkasse (bei der Bundesagentur für Arbeit)	Bund	Bundesministerium für Familie, Senioren, Frauen und Jugend (fachlich), Bundesministerium der Finanzen (steuerlich)
Elterngeld	Elterngeldstellen der Länder	Länder	Bundesministerium für Familie, Senioren, Frauen und Jugend
Unterhaltsvorschuss	Jugendämter	Kommune	Bundesministerium für Familie, Senioren, Frauen und Jugend

See also interesting [FAZ article](#) from today about the plan to harmonize different transfers.